



■ FX[®]-40HSE

Carthage Mills' FX-40HSE is a nonwoven geotextile designed specifically for the environmental market, is constructed of polypropylene staple fibers which are formed into a random network, needle-punched and heatset for dimensional stability. FX-40HSE is part of the Carthage [FX[®]-HSE Series](#) of nonwoven geotextiles, is inert to biological degradation and resistant to naturally encountered chemicals, alkalis, and acids.

The Carthage Mills' FX-HSE Series of environmental nonwoven geotextiles designates "Mass Per Unit Area" or Weight Per Square Yard" as a true Minimum value.

PROPERTY	TEST METHOD	DATA	
		METRIC	ENGLISH
☐ Mechanical			
Grab Tensile Strength	ASTM D 4632	0.47 kN	105 lbs
Grab Tensile Elongation		50%	
Trapezoidal Tear	ASTM D 4533	0.20 kN	45 lbs
CBR Puncture	ASTM D 6241	1.36 kN	305 lbs
☐ Endurance			
UV Resistance	ASTM D 4355	70% @ 500 hrs	
☐ Hydraulics / Filtration			
Permittivity ⁽¹⁾	ASTM D 4491	2.00 sec ⁻¹	
Water Flow Rate ⁽¹⁾		6518 l/min/m ²	160 g/min/ft ²
Apparent Opening Size (AOS) ⁽¹⁾	ASTM D 4751	0.212 mm	70 US Std. Sieve
☐ Physical			
Mass Per Unit Area (Minimum)	ASTM D 5261	135.6 g/m ²	4.0 oz/yd ²
Thickness (Typical)	ASTM D 5199	1.77 mm	70 mils
Standard Roll Sizes		4.58 m x 411.48 m	15 ft x 1350 ft

NOTES: Mullen Burst Strength ASTM D 3786 is no longer recognized by ASTM D35 on Geosynthetics. Puncture Strength ASTM D 4833 is not recognized by AASHTO M 288-06 and has been replaced with CBR Puncture ASTM D 6241.

⁽¹⁾ At the time of manufacturing. Handling, storage and shipping may change these properties.

- Unless otherwise stated, all values stated here are Minimum Average Roll Values (MARV).
- The properties reported above are effective 01-01-26 and are subject to change without notice.

Carthage Mills assumes no liability for the accuracy or completeness of this information or for the ultimate use by the purchaser. Carthage Mills disclaims any and all express, implied, or statutory standards, warranties or guarantees, including without limitation any implied warranty as to merchantability or fitness for a particular purpose or arising from a course of dealing or usage of trade as to any equipment, materials, or information furnished herewith. This document should not be construed as engineering advice.